HarbourOS

Port Management System

Technical Specification Document

Version 1.0 | April 2026

Classification: Confidential

1. Executive Summary

HarbourOS is a comprehensive port management platform designed to digitise and automate the full lifecycle of vessel port calls, container logistics, yard management, and associated billing. The system serves as the operational backbone of a port facility, replacing fragmented manual processes with a unified, event-driven platform.

The platform is built around a core insight: by the time a vessel physically arrives at berth, nearly all operational decisions should already be made. Pre-arrival workflows — manifest submission, container pre-declaration, slot allocation, and customs clearance — are completed days or weeks in advance. Physical arrival then triggers an execution phase tracked in real time against the pre-agreed plan.

HarbourOS serves multiple distinct user groups through dedicated portals: port operations staff, shipping agents, freight forwarders, and gate personnel. Each interface is purpose-built for its operational context, from desktop reconciliation dashboards to ruggedised tablet interfaces for dockside workers.

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2. System Scope

2.1 In Scope

- Vessel scheduling, berth allocation, and port call lifecycle management
- Pre-arrival manifest submission and automated reconciliation
- Container registry based on ISO 6346 standard
- Yard management — slot allocation, stacking rules, special handling
- Inbound and outbound container tracking
- Gate operations — truck check-in, container release authorisation
- Vessel services management — bunkering, water, provisions, pilotage
- Fee calculation and invoicing engine
- Exception and quarantine management
- Multi-portal access for different user roles
- Outbound manifest generation and cargo readiness reporting

2.2 Out of Scope

- Internal cargo manifests — what is inside a container at the item level
- Customs declaration processing — HarbourOS receives hold flags but does not process declarations
- Freight forwarder supply chain management
- Shipping line vessel scheduling beyond the port call window

- Trucking and last-mile delivery management

> *These out-of-scope areas represent natural candidates for future companion systems that would integrate with HarbourOS via API.*

3. Core Domain Model

3.1 Key Entities Overview

The following entities form the foundation of the HarbourOS data model. Each entity is described with its primary attributes and relationships.

3.1.1 Vessel

Represents a physical ship. Vessels are registered in the system independently of any specific port call.

IMO Number	Globally unique 7-digit vessel identifier (International Maritime Organization)
Vessel Name	Registered name of the vessel
Flag State	Country of registration
Vessel Type	Container ship, bulk carrier, ro-ro, etc.
Gross Tonnage	Used for port due calculations
LOA (Length Overall)	Determines berth length requirements
Beam	Width, determines berth compatibility
Max Draft	Determines tidal/depth constraints
Shipping Line	Operating company, links to Account entity
Port Agent	The agent company acting on behalf of the shipping line at this port

3.1.2 Port Call

A Port Call represents a single visit of a vessel to the port. It is the parent entity for all operations during that visit — berth allocations, manifests, container movements, services, and fees.

Port Call ID	System-generated unique identifier
Vessel	Reference to Vessel entity
Status	Expected / Arrived / In Progress / Completed / Cancelled

ETA	Estimated time of arrival, submitted by shipping agent
ATA	Actual time of arrival
ETD	Estimated time of departure
ATD	Actual time of departure
Purpose	Inbound cargo / Outbound cargo / Both / Transit / Services only
Pilotage Required	Boolean — most large vessels require harbour pilot
Total Fees	Computed aggregate of all fee events for this port call

3.1.3 Berth Visit

A port call may involve multiple berth visits — for example a vessel unloads, moves to anchorage while outbound cargo is prepared, then returns to berth for loading. Each physical docking is a separate Berth Visit record under the parent Port Call.

Berth Visit ID	System-generated identifier
Port Call	Parent port call reference
Berth	Allocated berth/dock position
Planned Arrival at Berth	Scheduled time
Actual Arrival at Berth	Recorded by port ops
Planned Departure from Berth	Scheduled time
Actual Departure from Berth	Recorded by port ops
Purpose	Unloading / Loading / Both / Services
Status	Scheduled / Active / Completed

3.1.4 Berth / Dock Position

A physical dock position within the port. Berths are a finite and highly valuable resource — their occupancy is the primary operational constraint the system manages.

Berth ID	Unique identifier
Name / Code	Human-readable berth reference e.g. Berth 7, Dock A
Length (metres)	Maximum vessel LOA that can be accommodated
Max Draft (metres)	Depth restriction

Crane Count	Number of ship-to-shore cranes serving this berth
Status	Available / Occupied / Maintenance / Reserved
Current Vessel	If occupied, reference to active berth visit

3.1.5 Container

The Container entity represents a physical shipping container as a registered asset. This record is independent of any port visit — a container exists in the registry whether it is currently in the yard, at sea, or in another country entirely.

Container ID	Full ISO 6346 code e.g. MSCU1234567 — validated on entry
Owner Code	3-character BIC registered prefix e.g. MSC
Serial Number	6-digit numeric serial
Check Digit	Computed and validated per ISO 6346 algorithm
Category Code	U = freight container, J = detachable equipment, Z = trailer
Size Code	20ft, 40ft, 45ft etc — ISO size type code
Type Code	Dry, Reefer, Open Top, Flat Rack, Tank etc
Tare Weight (kg)	Empty container weight
Max Gross Weight (kg)	Maximum permissible loaded weight
Manufacture Date	Date of manufacture
Registered Owner	Company that owns the container — links to Account

> *The Container registry grows organically as containers pass through the port. When a container ID appears in a pre-declaration or manifest that does not exist in the registry, the system prompts creation of the record. ISO 6346 check digit validation is performed at the point of entry.*

3.1.6 Container Visit

A Container Visit represents a single instance of a container passing through the port. One container may have many visits over its lifetime. This is the operational record that tracks a container's journey through the port from pre-declaration to gate-out.

Visit ID	System-generated identifier
Container	Reference to Container entity
Port Call (Inbound)	The vessel port call that brought this container in

Port Call (Outbound)	The vessel port call taking this container out, if applicable
Freight Forwarder	Account responsible for this container visit
Cargo Categories	One or more: Electronics, Clothing, Food, Machinery, Chemicals etc
Declared Weight (kg)	Gross weight as declared on manifest
Hazmat Class	UN hazmat class if applicable — drives slot allocation rules
Reefer Required	Boolean — requires powered slot
Oversize	Boolean — requires special handling and placement
Status	See Container Visit State Machine (Section 4.1)
Allocated Slot	Reference to Yard Slot entity
Storage Start	Timestamp when container placed in yard — billing starts
Storage End	Timestamp when container leaves yard — billing ends
Customs Status	Cleared / Hold / Pending
Release Authorisation	Reference to party authorised to collect

3.1.7 Yard Slot

Represents a physical storage position within the port yard. Slots are organised into rows and bays, and can be stacked to a defined maximum height.

Slot ID	Unique identifier e.g. A-04-R2-S3 (Bay A, Position 4, Row 2, Stack 3)
Bay	Yard zone or bay reference
Position	Position within bay
Row	Ground row number
Stack Level	Height in stack (1 = ground level)
Max Stack Height	Maximum containers that can be stacked at this position
Type	Standard / Reefer-powered / Hazmat-isolated / Oversize
Ownership	Owned (by account) / Available for rental
Owning Account	If owned, reference to Account — null if available
Status	Available / Reserved / Occupied / Maintenance

Current Container Visit	If occupied, reference to active container visit

3.1.8 Account

Represents any company that has a commercial relationship with the port. This includes freight forwarders, shipping lines, port agents, and companies that own yard slots.

Account ID	System-generated identifier
Company Name	Legal company name
Account Type	Freight Forwarder / Shipping Line / Port Agent / Slot Owner / Ad Hoc
Registration Number	Company registration / tax ID
Billing Address	Primary billing contact details
Credit Terms	Agreed payment terms — null for ad hoc accounts (pay upfront)
Status	Active / Suspended / Closed
Owned Slots	List of yard slot references permanently assigned to this account
Default Fee Schedule	Reference to applicable fee schedule

3.1.9 Manifest

Manifests represent the formal cargo documentation submitted at key points in the port call workflow. Multiple manifest records exist per port call, each representing a different stage or version of the cargo declaration.

Manifest ID	System-generated identifier
Port Call	Parent port call reference
Type	Inbound Planned / Inbound Reconciled / Inbound Actual / Outbound Planned / Cargo Readiness / Final Outbound
Submitted By	Account or user who submitted
Submitted At	Timestamp
Format	JSON / EDIFACT / XML / Manual Entry
Status	Draft / Submitted / Reconciled / Approved / Superseded
Container Entries	List of container references with visit details

3.1.10 Fee Event

Every billable action in the system generates a Fee Event. This approach provides a complete, auditable trail of all charges rather than calculating fees as a lump sum.

Fee Event ID	System-generated identifier
Port Call	Parent port call reference
Container Visit	Related container visit if applicable
Event Type	See Fee Schedule (Section 6)
Occurred At	Timestamp of the triggering action
Quantity	Units to be billed e.g. number of crane lifts, days of storage
Unit Rate	Rate at time of event from applicable fee schedule
Total Amount	Computed: quantity × unit rate
Invoiced	Boolean — whether included in a generated invoice
Invoice Reference	Reference to Invoice entity once raised

3.1.11 Exception Record

Any container or manifest entry that cannot be automatically processed generates an Exception Record for port operations staff to resolve.

Exception ID	System-generated identifier
Type	Invalid Container ID / Unmatched Manifest Entry / No Booking / Customs Hold / Damaged / Unmatched on Departure
Container Visit	Related container visit if applicable
Port Call	Related port call
Raised At	Timestamp
Raised By	System / User reference
Status	Open / In Progress / Resolved / Escalated
Resolution Notes	Free text audit trail of actions taken
Resolved By	User reference
Resolved At	Timestamp

4. State Machines

4.1 Container Visit States

Every container visit moves through a defined set of states. All fee triggers, slot allocations, and gate release rules are governed by these states.

State	Description	Key Triggers
Pre-declared	Freight forwarder has submitted booking. Container expected but not yet on a confirmed manifest.	Freight forwarder pre-declaration submitted
Manifested	Container appears on the captain's inbound manifest. Pre-declaration matched.	Manifest reconciliation — matched entry
Unmatched	Container on manifest with no pre-declaration, or pre-declaration with no manifest entry.	Reconciliation failure — exception raised
Confirmed	Manifest matched to pre-declaration. Slot allocated. Ready for unloading.	Reconciliation success + slot allocation complete
In Transit	Container physically being moved from vessel to yard slot.	Crane operation initiated
In Yard	Container placed in allocated yard slot. Storage billing begins.	Dockside worker confirms placement — timestamp recorded
On Hold	Customs or port authority hold applied. Cannot be released.	Customs hold flag received
Cleared	Customs hold lifted. Container eligible for release.	Customs clearance confirmed
Release Authorised	Port has received and approved release documentation.	Release order approved by port ops
Gate Out	Container has physically left the port via gate. Storage billing ends.	Gate staff confirm exit — timestamp recorded
Quarantine	Container has physical problem — invalid ID, damaged, or cannot be matched.	Exception raised by dockside or gate staff
Departed with Vessel	Container loaded onto outbound vessel and vessel has departed.	Final outbound manifest confirmed + ATD recorded

4.2 Port Call States

State	Description	Key Triggers
Scheduled	Agent has registered the port call. ETA submitted. Berth provisionally allocated.	Shipping agent pre-arrival notification
Manifest Submitted	Captain/agent has submitted inbound manifest. Reconciliation in progress.	Manifest upload — 24hrs before arrival
Reconciled	Manifest matched against all pre-declarations. Exceptions resolved or flagged.	Automated reconciliation complete
At Anchorage	Vessel has arrived in port waters but not yet berthed.	Pilot boat dispatched / AIS confirmation
Berthed	Vessel is at berth. Operations active.	Pilot confirms vessel secured
Operations Complete	All unloading/loading finished. Services may still be in progress.	Final container movement confirmed
Awaiting Outbound Cargo	Vessel moved to anchorage. Waiting for outbound containers to be ready.	Port ops instruction issued
Departed	Vessel has left the port. Final manifest generated. All fees closed.	Outbound pilot confirms vessel clear of port

5. Pre-Arrival Workflow

5.1 Overview

The pre-arrival workflow is the most critical phase of port operations. When executed correctly, a vessel's physical arrival is a logistics execution event — not a planning event. All decisions about what will be unloaded, where it will go, and what will be loaded should be resolved before the ship enters port waters.

5.2 Timeline

Timeframe	Actor	Action	System Response
3-6 weeks before	Freight Forwarder	Submits container pre-declaration: vessel,	Creates container visit records in Pre-

		ETA, container count, storage duration, cargo categories, special handling flags	declared state. Runs capacity check. Reserves yard slots provisionally. Notifies if capacity is insufficient.
1-2 weeks before	Shipping Agent	Registers port call. Requests berth allocation. Submits vessel details and estimated dwell time.	Creates Port Call record. Allocates berth based on vessel dimensions and schedule. Calculates estimated port dues.
48 hours before	Port Ops	Reviews capacity dashboard. Confirms all pre-declarations have storage arrangements. Flags unresolved exceptions to freight forwarders.	Generates pre-arrival readiness report. Lists containers with incomplete arrangements.
24 hours before	Shipping Agent / Captain	Submits formal inbound manifest via EDI, XML, or JSON. Declares all containers on board and destined for this port.	Runs automated reconciliation against all pre-declarations. Generates reconciliation report. Raises exceptions for unmatched entries. Advances matched containers to Confirmed state.
12 hours before	Port Ops	Reviews and resolves reconciliation exceptions. Approves or rejects unmatched containers. Finalises slot allocation plan.	Updates container visit states. Notifies freight forwarders of unresolved issues. Generates crane operation sequence plan.
On arrival	Harbour Pilot	Boards vessel. Guides vessel to allocated berth.	Records ATA. Initiates berth visit record. Activates operations phase.

5.3 Capacity Management

HarbourOS maintains a forward-looking capacity model. At any point, port operations can view projected yard occupancy over the next 96 hours based on all confirmed incoming vessels and expected gate-out activity. This enables proactive management of capacity constraints before they become operational crises.

When a pre-declaration or manifest submission would exceed yard capacity, the system responds as follows:

- Immediate capacity check on submission — not deferred
- Partial acceptance — slots confirmed for containers that fit, flagged for those that do not
- Priority enforcement — owned slots are guaranteed regardless of overall capacity; spot-rental containers are at risk
- Direct delivery option — containers with confirmed immediate collection can be flagged as gate-through, bypassing yard storage entirely
- Alert to port ops — capacity breach flagged for human decision on anchorage, transhipment or prioritisation

6. Manifest Reconciliation

6.1 Reconciliation States

When an inbound manifest is received, every container entry is cross-referenced against existing pre-declarations. Each entry lands in one of four reconciliation states:

Matched and Confirmed	Pre-declaration exists. Storage arranged. Container advances to Confirmed state. No action required.
Matched — Incomplete	Pre-declaration exists but storage not yet arranged. Exception raised. Freight forwarder notified urgently. Container cannot be unloaded until resolved.
On Manifest — No Booking	Container appears on manifest with no corresponding pre-declaration. Exception raised. Port ops must approve before container can be unloaded. Default position is to leave on vessel.
Booked — Not on Manifest	Pre-declaration exists but container does not appear on manifest. Cargo missed the vessel. Exception raised. Reserved slot released back to available inventory. Freight forwarder notified.

6.2 Manifest Formats

HarbourOS accepts manifests in multiple formats. All formats are parsed into a canonical internal manifest model before reconciliation. The parsing layer is separate from the reconciliation logic — adding new format support does not require changes to the reconciliation engine.

Format	Standard	Notes
JSON API	HarbourOS native schema	Primary format for modern integrations. Schema published in API documentation.
EDIFACT	UN/EDIFACT — COPRAR/COARRI messages	Legacy format used by major shipping lines. Supported via dedicated parser.
XML	Various port community schemas	Accepted with format auto-detection. Falls back to manual field mapping if schema unknown.
Manual Entry	N/A	Port ops staff can enter manifest data directly via admin portal. Full audit trail recorded.

6.3 Container ID Validation

All container IDs are validated against the ISO 6346 standard at the point of entry regardless of submission format. Validation includes:

- Format check — owner code (3 alpha), category (U/J/Z), serial number (6 numeric), check digit (1 numeric)
- Check digit algorithm — mathematically computed and verified
- Owner code lookup — cross-referenced against BIC registered prefix database
- Registry lookup — if container exists in registry, declared properties validated against registered properties

> *A container ID that fails check digit validation is not rejected outright at the dockside — the physical container must still be handled. Instead it is placed in quarantine and an exception record is raised. The invalid ID is logged alongside the physical description for manual resolution.*

7. Yard Management

7.1 Slot Allocation Rules

Slot allocation is performed automatically by the system when a container visit reaches Confirmed state. The allocation engine applies the following rules in priority order:

- Owned slots — containers belonging to accounts with permanently owned slots are allocated there first

- Special handling requirements — reefer containers must be in powered slots; hazmat containers must be in designated hazmat zones with required separation distances; oversize containers require appropriately dimensioned ground positions
- Proximity grouping — containers from the same freight forwarder or same vessel call are preferentially allocated in adjacent positions to minimise crane and vehicle travel
- Stack optimisation — heavier containers on lower stack levels; containers with earliest expected departure on top of stacks
- Rental allocation — remaining containers are allocated from available rental slots in proximity groups

7.2 Special Handling Categories

Reefer	Requires powered slot with electrical connection. Temperature monitoring required. Slot count is a hard constraint — system will flag if insufficient powered slots available.
Hazmat	UN hazmat class must be declared. Allocated to designated hazmat zone. Separation distances from other hazmat containers enforced per IMDG code class compatibility matrix.
Oversize	Requires ground-level placement. May occupy multiple slot footprints. Crane and vehicle routing must be adjusted. Port ops notified for manual review of placement.
High Value	Electronics and other high-value cargo can be flagged for placement in secured yard zones where available.
Damaged	Containers recorded as damaged on arrival are placed in a designated inspection area. Cannot be released until surveyor inspection is complete and sign-off recorded.

7.3 Quarantine Zone

A dedicated area of the yard is designated as the quarantine zone. Containers are placed here when they cannot be processed through the normal workflow. Quarantine placement does not block yard operations — the container is handled physically but its records are flagged for resolution.

Containers may enter quarantine for the following reasons:

- Invalid or unreadable container ID
- No matching booking or pre-declaration

- Physical damage preventing normal storage
- Customs seizure or authority hold requiring segregation
- Contents mismatch discovered during inspection

8. Vessel Services

8.1 Services Overview

HarbourOS manages the scheduling, fulfilment, and billing of all services provided to vessels during a port call. Services are initiated by the shipping agent during the pre-arrival phase and can be added or modified during the port call.

Service	Description	Scheduled By	Billing Trigger
Pilotage (Inbound)	Harbour pilot boards vessel and guides it to berth	Auto-scheduled on berth confirmation	Completion of inbound pilotage
Pilotage (Outbound)	Harbour pilot guides vessel out of port	Auto-scheduled on ETD confirmation	Completion of outbound pilotage
Bunkering	Fuel supply to vessel via bunker barge or truck	Shipping agent request	Delivery quantity confirmed
Fresh Water	Potable water supply	Shipping agent request	Delivery quantity confirmed
Provisions	Food and crew supplies — port facilitates dockside access for suppliers	Shipping agent request	Access time and facilitation fee
Waste Disposal	Regulated discharge of vessel waste — legally required	Mandatory — auto-raised on port call creation	Volume discharged
Crew Services	Crew change facilitation, shore leave coordination	Shipping agent request	Per crew member or flat fee
Maintenance Access	Facilitation of contractor dockside access for vessel repairs	Shipping agent request	Access duration and contractor count

8.2 Berth State and Anchorage Management

HarbourOS actively manages the movement of vessels between berth and anchorage to optimise berth utilisation. When a vessel completes unloading but its outbound cargo is not yet ready, the system evaluates whether the berth is needed for another vessel and, if so, issues an anchorage instruction to the shipping agent.

The system tracks each vessel in one of the following location states:

- At sea — vessel en route, not yet in port waters
- At anchorage — vessel in port anchorage area, awaiting berth or awaiting cargo
- At berth — vessel physically docked, operations active or services in progress
- Departed — vessel has left port waters

> *Anchorage time is billable at a lower rate than berth time but is still recorded and invoiced. The system automatically calculates time in each location state from the port call event log.*

9. Gate Operations

9.1 Truck Inbound — Container Drop-off

When a truck arrives at the gate to deliver a container for loading onto a vessel, the gate workflow is:

- Gate staff scan or enter truck registration and driver ID
- Driver presents delivery order — gate staff verify against expected outbound container records
- Container ID scanned — validated against outbound manifest booking
- Seal number recorded and matched against manifest declaration
- Container inspected for damage before acceptance
- Gate-in timestamp recorded — container enters port custody
- System allocates yard slot and provides routing instructions to truck driver

9.2 Truck Outbound — Container Collection

When a truck arrives to collect a container from the yard, the gate workflow is:

- Gate staff scan truck registration and driver ID
- Collection order presented — gate staff verify release authorisation in system
- System checks: container is in Cleared state, release authorisation is valid, collecting party matches authorised party
- If any check fails — collection is refused and exception is raised
- If all checks pass — gate-out is approved, container location in yard is provided to yard vehicle
- Container physically moves from yard to gate

- Gate staff confirm exit — gate-out timestamp recorded, storage billing ends

> *A container in On Hold state cannot be released under any circumstances until the hold is formally lifted and the status updated by an authorised user. Gate staff have no ability to override a hold — they can only see that collection is blocked and contact port ops.*

10. Fee Schedule and Billing

10.1 Fee Categories

All fees in HarbourOS are event-driven. A Fee Event record is created at the moment the triggering action occurs, with the rate applicable at that time. Invoices are generated by aggregating fee events within a billing period or at port call completion.

Fee Category	Triggering Event	Unit	Notes
Port Dues	Vessel arrival	Per gross tonne	Based on vessel gross tonnage. Charged once per port call.
Berth Hire	Time at berth	Per hour or part thereof	Clock starts when vessel secured, stops when vessel departs berth. Anchorage is a separate lower rate.
Anchorage	Time at anchorage	Per hour or part thereof	Lower rate than berth hire.
Pilotage	Pilot service completion	Per movement (in/out)	Fixed fee per pilotage movement. Inbound and outbound billed separately.
Crane / Lift Fee	Each container crane movement	Per lift	Charged per container movement, both on and off vessel.
Storage	Container in yard	Per container per day or part thereof	Clock starts on yard placement, stops on gate-out. Reefer storage at premium rate.
Gate Fee	Each truck gate transaction	Per gate movement	Inbound and outbound truck movements.
Bunkering Facilitation	Fuel delivery completion	Per tonne delivered or flat fee	Port facilitates, charges for service.
Water Supply	Delivery completion	Per tonne	

Waste Disposal	Discharge completion	Per cubic metre by waste type	Regulated — mandatory service.
Overtime	Operations outside standard hours	Multiplier on applicable fees	Operations requested outside standard port operating hours attract a surcharge.

10.2 Invoicing

Invoices are generated per port call on completion, or at agreed billing cycle intervals for accounts with standing credit arrangements. Each invoice references the underlying fee events providing full transparency and auditability. Ad hoc accounts with no credit terms are billed upfront based on estimated fees, with a reconciliation invoice issued on port call completion.

11. User Roles and Portals

11.1 Portal Overview

Portal	Primary Users	Core Functions
Port Operations Dashboard	Port ops staff, harbour masters	Live berth and yard occupancy, incoming vessel schedule, exception management, manifest reconciliation, vessel service coordination, capacity forecasting
Shipping Agent Portal	Vessel agents, shipping lines	Port call registration, manifest submission, berth requests, vessel service requests, outbound container status, fee estimates
Freight Forwarder Portal	Freight forwarders, customs brokers	Container pre-declaration, slot booking, container visit tracking, release authorisation management, invoice viewing
Gate Terminal	Gate staff	Truck check-in, container verification, release authorisation check, gate-in and gate-out recording — optimised for speed with simple UI
Dockside Tablet	Dockside workers, crane operators	Container ID confirmation, slot assignment display, placement

		confirmation, exception flagging — ruggedised UI, offline tolerant
Finance and Admin	Finance staff, port management	Fee schedule management, invoice generation, account management, revenue reporting, audit logs

11.2 Dockside Interface Considerations

The dockside tablet interface has specific requirements that distinguish it from other portals:

- Large touch targets — workers wearing gloves in outdoor environments
- High contrast display — readable in direct sunlight
- Minimal input — container ID entry or barcode/QR scan, then confirmation
- Offline tolerance — operations queue locally if yard WiFi is interrupted and sync on reconnection
- Immediate feedback — clear visual and optionally audio confirmation on successful slot assignment
- Exception escalation — simple one-tap flagging to raise an exception to port ops without interrupting workflow

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12. Technology Stack

12.1 Backend

Runtime	Node.js
Framework	AdonisJS v6
Language	TypeScript
ORM	Lucid ORM (AdonisJS native)
Database	PostgreSQL
Authentication	AdonisJS Auth — session-based for portals, token-based for API
Job Queue	BullMQ — for async processing of manifest reconciliation, fee calculations, notifications
Real-time	WebSockets via AdonisJS Transmit — for live berth and yard dashboards

12.2 Frontend

Template Engine	Edge.js (AdonisJS native)
UI Framework / Base	TailAdmin — Tailwind CSS v4 admin template
CSS Framework	Tailwind CSS v4
JavaScript	Alpine.js v3 — lightweight reactivity for UI components
Build Tool	Vite with @adonisjs/vite integration
Charts	ApexCharts — operational dashboards and reports
Calendar	FullCalendar — vessel scheduling and berth calendar views
Date Picker	Flatpickr
Package Management	npm

12.3 Infrastructure Considerations

- The dockside tablet interface must degrade gracefully under poor connectivity — service worker caching and local action queuing required
- The operations dashboard requires real-time updates — WebSocket connection to push berth status, yard changes, and exception alerts without page refresh
- Manifest file processing (particularly EDIFACT parsing) should be handled asynchronously via job queue — large manifests should not block the HTTP request cycle
- All user actions on container records, manifest approvals, and exception resolutions must be written to an immutable audit log

13. External Integration Points

13.1 Inbound Integrations

Manifest Submission	EDIFACT (COPRAR/COARRI), XML, JSON — received from shipping lines and agents
AIS (Automatic Identification System)	Vessel position and ETA updates from maritime AIS feeds — used to track vessel approach and update ETA automatically
Customs Authority	Hold and clearance status notifications from national customs systems
BIC Prefix Registry	Owner code validation and lookup — queried on container registration

Payment Gateway	For ad hoc account upfront payment processing

13.2 Outbound Integrations

Manifest Generation	Final outbound manifests generated in JSON and EDIFACT for onward transmission to destination ports
Customs Notification	Inbound cargo notifications to national customs authority on manifest receipt
Invoice Delivery	PDF invoices generated and delivered to account billing contacts
Notifications	Email and optionally SMS alerts to freight forwarders, agents, and gate staff for exception alerts, cargo readiness, and collection notifications

13.3 Future Integration Candidates

The following systems represent natural future integration points as the platform matures, and should be considered when designing the API layer:

- Freight forwarder platform — pre-declarations submitted directly via API rather than through the portal
- Trucking / haulage management systems — gate appointments and collection scheduling
- Port community systems — regional port data sharing platforms
- Banking and trade finance systems — bill of lading and letter of credit workflows

14. Suggested Development Phases

Phase 1 — Core Foundation

- Database schema and entity model
- Account and user management with role-based access
- Container registry with ISO 6346 validation
- Vessel and berth management
- Basic port call lifecycle
- Port operations dashboard — berth view

Phase 2 — Pre-Arrival Workflow

- Freight forwarder portal — container pre-declaration
- Shipping agent portal — port call registration and manifest submission
- Manifest reconciliation engine
- Yard slot allocation engine
- Capacity management and forecasting
- Exception management

Phase 3 — Operational Execution

- Dockside tablet interface with offline support
- Gate terminal interface
- Container visit state machine — full lifecycle
- Vessel services scheduling and fulfilment
- Real-time dashboard updates via WebSocket

Phase 4 — Billing and Reporting

- Fee event engine — all billable actions
- Invoice generation
- Finance and admin portal
- Operational reports and analytics
- Audit log interface

Phase 5 — Integrations and Polish

- EDIFACT parser for inbound manifests
- AIS feed integration for vessel ETA tracking
- Outbound manifest generation in EDIFACT and JSON
- Customs notification integration
- API documentation for future third-party integrations

Appendix A — ISO 6346 Container ID Format

Every shipping container has a globally unique identifier conforming to ISO 6346. The format is:

Component	Length	Values	Example
Owner Code	3 chars	A-Z only. BIC registered prefix assigned to container owner.	MSC
Category Identifier	1 char	U = freight container, J = detachable equipment, Z = trailer/chassis	U
Serial Number	6 digits	Numeric 000000–999999. Assigned by owner. Up to 1,000,000 unique containers per owner code.	123456
Check Digit	1 digit	Computed via ISO 6346 algorithm from preceding 10 characters. Validates the entire ID.	7

Full example: MSCU1234567

With 17,576 possible three-character owner codes (26^3) and 1,000,000 serial numbers per owner code, the standard supports up to 17.576 billion unique container identifiers — far exceeding the estimated 50 million containers currently in existence worldwide.

Appendix B — Glossary

AIS	Automatic Identification System — vessel tracking system broadcasting position and ETA
ATA / ATD	Actual Time of Arrival / Actual Time of Departure
BIC	Bureau International des Containers — registers and licenses container owner prefix codes
Bunkering	Fuelling a vessel at port
EDIFACT	UN/EDIFACT — international EDI standard used for cargo messaging in the shipping industry
ETA / ETD	Estimated Time of Arrival / Estimated Time of Departure
Freight Forwarder	A company that arranges cargo shipments on behalf of cargo owners. The port's primary commercial counterparty for container operations.
HAZMAT	Hazardous materials — containers carrying dangerous goods classified under the IMDG code

IMO Number	International Maritime Organization unique vessel identifier — 7 digits, permanent for the life of the vessel
ISO 6346	International standard for shipping container identification and marking
LCL	Less than Container Load — consolidated shipment where multiple shippers share one container
Manifest	Official cargo document declaring all goods being carried on a vessel
NOA	Notice of Arrival — formal pre-arrival notification submitted by shipping agent to port authority
Port Agent	A company acting on behalf of a shipping line at a specific port — handles all port-side arrangements for the vessel
Port Call	A single visit of a vessel to a port, from arrival to departure
Reefer	Refrigerated container requiring temperature control and electrical power connection
Transhipment	Transfer of containers from one vessel to another at an intermediate port
UN/LOCODE	United Nations Code for Trade and Transport Locations — standardised port and location codes